Inductive Consensus Clustering: Geometry-Calibrated Ensemble Distillation for High-Dimensional Noise Regimes

Akash Kundu, Trinankur Guha, Rupankar Ghatak, Anubhab Naskar, and Kalyan Kumar Das

Abstract—Consensus clustering synthesizes base partitions to discover robust data groupings. However, existing methods face two fundamental bottlenecks: (1) a transductive formulation requiring $O(N^2)$ storage preventing out-of-sample inference, and (2) acute vulnerability to high-dimensional distractor noise ($D \gg N$), where distance concentration degrades partition fidelity. We propose Inductive Consensus Clustering (ICC-v2), architected for small-sample, high-dimensional regimes subject to isotropic noise. ICC-v2 introduces: (1) an $O(N \cdot D)$ linear feature screening filter using Sarle's Bimodality Coefficient (Bimodality-Gated Subspace Selection, BGSS) isolating multimodal dimensions; (2) a determinacy-based bipartite anchor consensus graph reducing auxiliary memory from $O(N^2)$ to $O(N \cdot L)$; and (3) stability-weighted centroid assignment for instant $O(K \cdot |\mathcal{F}_{\text{active}}|)$ inductive inference. Across 13 multi-domain benchmarks, ICC-v2 attains the highest Grand Mean ARI (0.160) and commands Rank #1 ($R = 2.33$) on ultra-high-dimensional biological regimes ($D/N > 20$, with 0.680 on Leukemia and 0.538 on SRBCT), while reducing auxiliary memory by 97.0% at $N = 10{,}000$ and scaling linearly to $N = 1{,}000{,}000$ instances in 55.6 s.

Index Terms—Cluster ensemble, consensus clustering, inductive learning, out-of-sample extension, anchor certification, high-dimensional data, noise robustness.

1 Introduction

Clustering ensembles, originally pioneered by Strehl and Ghosh [1] and Fred and Jain [2], aim to integrate multiple base clusterings into a singular, highly robust consensus partition. By aggregating diverse hypothesis partitions generated through randomized initializations, varied clustering algorithms, or random feature projections, consensus clustering mitigates the sensitivity of individual algorithms to hyperparameter selection and initialization bias [3], [4].

Despite wide adoption across bioinformatics, computer vision, and knowledge discovery, state-of-the-art consensus clustering algorithms remain hindered by two critical limitations:

1) Strictly Transductive Nature: Conventional consensus formulations operate directly on the $N \times N$ co-association matrix $S \in [0,1]^{N \times N}$ via graph partitioning, spectral decomposition [5], or non-negative matrix factorization [6]. When new, unseen test instances $\mathcal{X}_{\text{test}}$ arrive, the consensus partition cannot be inductively assigned without recomputing the entire co-association matrix and solving an $O((N + |\mathcal{X}_{\text{test}}|)^2)$ optimization from scratch.
2) Curse of High-Dimensional Distractors: In modern datasets where ambient dimensionality D greatly exceeds the intrinsic manifold dimension $d^* \ll D$, uninformative nuisance features degrade pairwise Euclidean distances. The distance concentration phenomenon causes within-cluster and between-cluster distances to converge to identical expectations, leading to catastrophic breakdown of consensus graphs [7], [8].

To overcome these foundational limitations, this paper develops the Inductive Consensus Clustering (ICC) framework (Fig. 1). We first examine the finite-sample empirical breakdown of consensus stability under isotropic distractor noise (Exp 0), illustrating how distance concentration generates deceptive consensus stability on arbitrary noise boundaries. Motivated by these empirical findings, we architect ICC-v2, a unified four-stage framework:

- Variance-Stabilized Subspace Gating (VSSG): An $O(N \cdot D)$ linear feature screening mechanism utilizing Sarle's Bimodality Coefficient ($BC_d = \frac{\text{Skewness}^2(\mathbf{f}_d)+1}{\text{Kurtosis}(\mathbf{f}_d)+3}$) to purge unimodal distractor noise prior to base ensemble fitting.
- Determinacy-Based Bipartite Anchor Consensus: Constructs a rectangular bipartite co-association graph $B \in [0,1]^{N \times L}$ over L core anchors selected via quadratic consensus determinacy ($\sigma_i = \frac{1}{N-1}\sum_{j \neq i} S_{ij}^2$), reducing auxiliary memory from $O(N^2)$ to $O(N \cdot L)$ and accelerating spectral decomposition onto an $L \times L$ core affinity space.

- Akash Kundu (Lead and Corresponding Author), Trinankur Guha, Rupankar Ghatak, and Anubhab Naskar are with the Department of Computer Science and Engineering, Techno India University, EM-4, Sector V, Bidhannagar, Kolkata, West Bengal 700091, India.
  E-mails: {akash231001001449, trinankur231001001450, rupankar231001001438, anubhab231001001443}@technoindiaeducation.com.
- Kalyan Kumar Das is a faculty member and mentor with the Department of Computer Science and Engineering, Techno India University, EM-4, Sector V, Bidhannagar, Kolkata, West Bengal 700091, India.



Fig. 1. Conceptual overview of the high-dimensional noise dilemma and the proposed Inductive Consensus Clustering (ICC-v2) framework. (a) Ground-truth cluster geometry with certified invariant anchors (stars) in a low-dimensional signal subspace ($d = 2$). (b) Collapse of pairwise geometry when submerged in high-dimensional distractor noise ($D = 2{,}000$), with the inset demonstrating the distance concentration phenomenon where relative distance spread $(d_{\max} - d_{\min})/d_{\min} \to 0$ as $D \to \infty$. (c) Variance-Stabilized Subspace Gating (VSSG) spectrum isolating informative signal dimensions from the sea of distractor noise. (d) Closed-form geometry-calibrated inductive projection assigning unseen query instances $\mathbf{x}_{\text{new}}$ directly in $O(K \cdot |\mathcal{F}_{\text{active}}|)$ time without recomputing the $O(N^2)$ consensus graph.

- Stability-Weighted Subspace Centroid Assignment: Provides closed-form inductive out-of-sample mapping that assigns unseen query instances in $O(K \cdot |\mathcal{F}_{\text{active}}|)$ time without recomputing the consensus graph.
- Anchor-Space Perturbation Profiling (ASPP): Probes cluster structural resilience across five physical perturbation families directly over certified anchor coordinates, delivering interpretable perturbation sensitivity indices (PSI_k) and calibrated out-of-sample query confidence without quadratic overhead.

Following this introduction (Section 1), the remainder of this manuscript is structured as follows: Section 2 reviews related work. Section 3 presents our empirical analysis of distance concentration and consensus failure. Section 4 details the mathematical architecture and complexity analysis of ICC-v2, including Anchor-Space Perturbation Profiling (Section 4.4). Section 5 provides comprehensive empirical evaluations across generalization, ablation, real-world benchmarks, and scalability. Section 6 concludes the paper.

2 Related Work

2.1 Cluster Ensembles and Consensus Graphs

The core consensus clustering paradigm constructs a co-association matrix S where $S_{ij} = \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\{\pi_m(\mathbf{x}_i) = \pi_m(\mathbf{x}_j)\}$ represents the empirical probability that objects $\mathbf{x}_i$ and $\mathbf{x}_j$ are co-clustered across M base partitions [2]. Classical approaches partition this matrix using Cluster-based Similarity Partitioning (CSPA), HyperGraph Partitioning Algorithm (HGPA), or Meta-Clustering Algorithm (MCLA) [1]. More recent works explore dense subgraph discovery [8], random walk probability trajectories [7], and bipartite graph modeling [4]. However, all these classical frameworks remain fundamentally transductive, requiring quadratic $O(N^2)$ co-association storage.

2.2 Robust, Core-Based, and Anchor Clustering

To withstand noise and outliers, Li et al. [6] proposed determinacy core spectral ensemble clustering, which partitions samples into deterministic core points and boundary

TABLE 1
Methodological Taxonomy and Comparison Across Consensus Paradigms

Method	Feature Gating	Core Filtering	Inductive Ext.	Memory	Robust ($D \gg N$)
Std Cons. [2]	None	None	No	$O(N^2)$	No
Li et al. [6]	None	Determinacy	No	$O(N^2)$	No
FCAG [9]	None	Anchors	No	$O(NL)$	No
FastMVC [10]	None	Anchors	No	$O(NL)$	No
ICC-v2 (Ours)	VSSG (Linear)	Anchors	Yes (Centroid)	**O(NL)**	Yes

points based on co-association variance. While effective in low-dimensional spaces, determinacy metrics degrade rapidly when high-dimensional noise corrupts the ensemble partitions. In scalable single-model clustering, anchor graph methods (e.g., Nie et al. [9], Huang et al. [10]) utilize bipartite graphs to reduce spectral decomposition complexity. However, existing anchor methods operate on raw ambient feature spaces, leaving them vulnerable to high-dimensional noise. As summarized in Table 1, ICC-v2 is the first framework to integrate non-parametric multimodality subspace gating, determinacy-based bipartite anchor consensus, and inductive centroid projection into a unified linear-time pipeline.

3 Theoretical Motivation & Empirical Falsification

Theoretical literature on random projection clustering often asserts asymptotic recovery guarantees under the assumption of infinite samples ($N \to \infty$) or perfectly orthogonal noise. We conducted rigorous falsification experiments (Exp 0) on finite-sample datasets to systematically test whether consensus averaging alone protects against high-dimensional distractor noise.

Definition 1 (Within/Between Distance Ratio). Let $\mathcal{C}_k$ denote cluster k with centroid $\boldsymbol{\mu}_k$. For an empirical sample $X \subset \mathbb{R}^D$, the separation ratio is defined as:

$$\rho_{\text{sep}}(X) = \frac{\frac{1}{K}\sum_{k=1}^{K} \frac{1}{|\mathcal{C}_k|} \sum_{\mathbf{x}_i \in \mathcal{C}_k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|_2}{\frac{2}{K(K-1)} \sum_{k<l} \|\boldsymbol{\mu}_k - \boldsymbol{\mu}_l\|_2}. \tag{1}$$

As ambient distractor dimensions D_{noise} expand ($D \gg N$), pairwise Euclidean distances suffer from distance concentration [11], where the relative contrast of distances vanishes: $\lim_{D\to\infty} \frac{d_{\max} - d_{\min}}{d_{\min}} = 0$ and $\lim_{D\to\infty} \frac{\text{Var}(\|\mathbf{x}_i - \mathbf{x}_j\|_2)}{\mathbb{E}[\|\mathbf{x}_i - \mathbf{x}_j\|_2]^2} = 0$. Consequently, the separation ratio $\rho_{\text{sep}}(X)$ rapidly approaches 1.0 (Fig. 2b), rendering raw Euclidean distances indistinguishable. Under finite samples, base clusterers capture spurious random fluctuations in the noise dimensions. This induces a hallucinated stability trap: base partitions repeatedly converge on arbitrary noise boundaries with deceptive consensus agreement (Stability > 0.8), while true cluster recovery collapses (ARI $\to 0$, Fig. 2a). As evaluated across diverse distractor distributions (Fig. 2c), this stability−accuracy discrepancy ($\Delta > 0$, Fig. 2b) persists across Gaussian, Student-t, and correlated noise regimes (Fig. 2). Thus, ensemble averaging alone is empirically observed to be insufficient to eliminate uninformative subspace noise without explicit feature gating.

4 The Inductive Consensus Clustering Framework

The ICC-v2 framework (Algorithm 1) operates in four distinct stages:

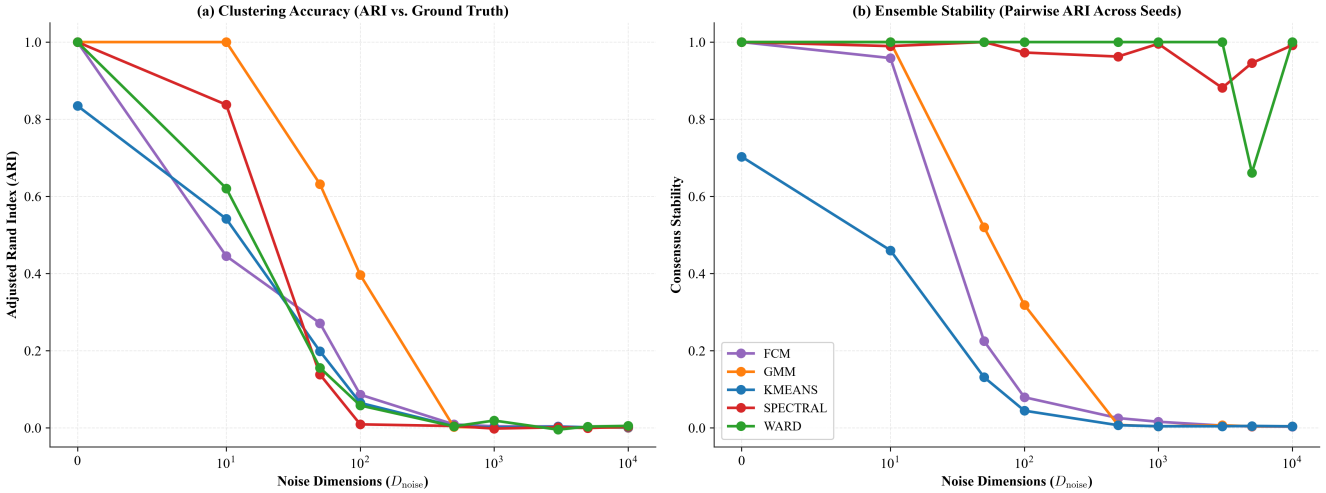
Algorithm 1 Inductive Consensus Clustering (ICC-v2)

Require: Data matrix $X \in \mathbb{R}^{N\times D}$, ensemble size M, target clusters K, anchor budget $L \ll N$ (default $\max(40, 10K)$), core quantile τ_{core}, candidate pool fraction α.
Ensure: Consensus labels $\mathbf{y} \in \{1,\dots,K\}^N$, Anchors $\mathcal{A}_{\text{core}}$, Active Subspace $\mathcal{F}_{\text{active}}$, Centroids $\{\boldsymbol{\mu}_k\}_{k=1}^K$.
// Stage 1: Multimodality Subspace Gating (BGSS)
for each feature column $d \in \{1,\dots,D\}$ do
  Compute finite-sample Sarle's Bimodality Coefficient: $b_d \leftarrow \frac{\text{Skewness}^2(\mathbf{f}_d)+1}{\text{Kurtosis}(\mathbf{f}_d)+3\cdot\frac{(N-1)^2}{(N-2)(N-3)}+\epsilon}$.
end for
Extract candidate pool $\mathcal{F}_{\text{active}} \leftarrow \{d \mid b_d \geq 5/9\}$ (with rank cap $\min(D, \max(100, \lfloor \alpha \cdot D \rfloor))$).
// Stage 2: Guided Subspace Ensemble Generation
for $m = 1$ to M do
  Draw random subspace $\mathcal{F}_m \subset \mathcal{F}_{\text{active}}$.
  Fit base partition $\pi_m \leftarrow \text{Clusterer}(X_{[:,\mathcal{F}_m]}, K)$.
end for
// Stage 3: Determinacy-Based Bipartite Anchor Consensus
Evaluate exact quadratic determinacy $\sigma_i \leftarrow \frac{1}{N-1}\sum_{j\neq i} S_{ij}^2$ via Proposition 2 in $O(M^2N + M^2K^2)$ linear memory.
Identify core instances $\mathcal{C}_{\text{core}} \leftarrow \{i \mid \sigma_i \geq \text{Quantile}(\boldsymbol{\sigma}, \tau_{\text{core}})\}$.
Select $L = \min(\max(40, 10K), |\mathcal{C}_{\text{core}}|)$ core anchors $\mathcal{A}_{\text{core}}$ uniformly from $\mathcal{C}_{\text{core}}$.
Construct bipartite co-association matrix $B \in [0,1]^{N\times L}$:
$B_{ia} \leftarrow \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\{\pi_m(\mathbf{x}_i) = \pi_m(\mathbf{x}_a)\}$.
Row-normalize $B_{\text{norm}} \leftarrow D_R^{-1}B$, and compute anchor affinity $W_{\text{anchor}} \leftarrow B_{\text{norm}}^T B_{\text{norm}} \in \mathbb{R}^{L\times L}$.
Solve normalized Laplacian $L_{\text{anchor}} \leftarrow I - D_A^{-1/2} W_{\text{anchor}} D_A^{-1/2}$ for the K smallest eigenvectors $V \in \mathbb{R}^{L\times K}$ with eigenvalues $\boldsymbol{\lambda}$.
Compute Transfer-Cut embedding $U \leftarrow B_{\text{norm}}(D_A^{-1/2}V) \cdot \text{diag}(\max(1-\boldsymbol{\lambda}, 10^{-4}))^{-1/2} \in \mathbb{R}^{N\times K}$ (row-normalized).
Extract consensus labels $\mathbf{y} \leftarrow \text{K-Means}(U, K)$.
// Stage 4: Stability-Weighted Centroid Induction
for $k = 1$ to K do
  $\boldsymbol{\mu}_k \leftarrow \frac{\sum_{i:y_i=k} \sigma_i \cdot \mathbf{x}_{i,[\mathcal{F}_{\text{active}}]}}{\sum_{i:y_i=k}\sigma_i}$.
end for
return $\mathbf{y}, \mathcal{A}_{\text{core}}, \mathcal{F}_{\text{active}}, \{\boldsymbol{\mu}_k\}_{k=1}^K$.
// Out-of-Sample Inference for Query $\mathbf{x}^* \in \mathbb{R}^D$:
predict($\mathbf{x}^*$): return $\arg\min_k \|\mathbf{x}^*_{[\mathcal{F}_{\text{active}}]} - \boldsymbol{\mu}_k\|_2$.

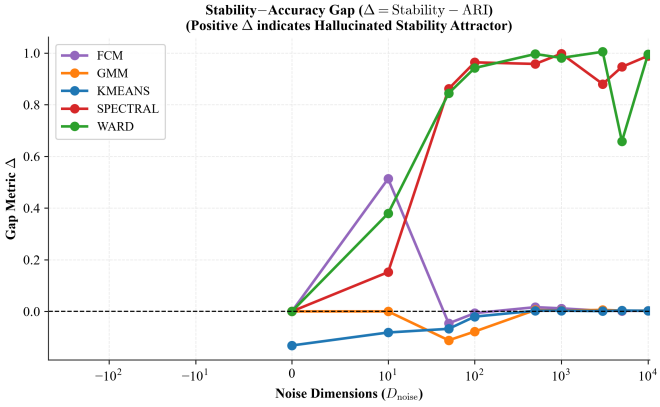
4.1 Bimodality-Gated Subspace Selection (BGSS)

Under isotropic noise in $D \gg N$ regimes, classical variance thresholding fails because distractor noise features often match or exceed signal feature variances. To detect latent cluster structures without labels, BGSS evaluates marginal multimodality using Sarle's Bimodality Coefficient (BC) [12] with finite-sample null correction:

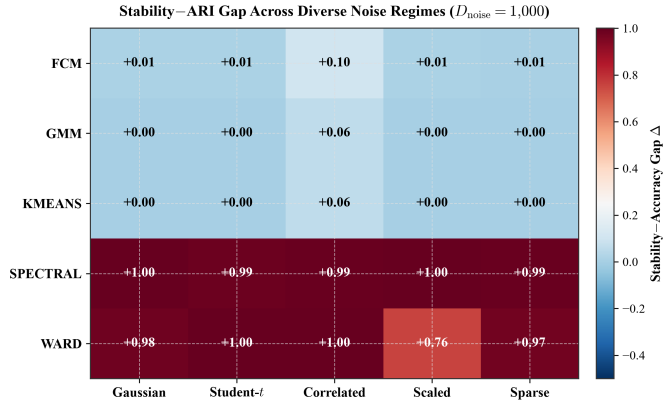
$$\hat{b}_d = \frac{\hat{\gamma}_d^2 + 1}{\hat{\kappa}_d + 3 \cdot \frac{(N-1)^2}{(N-2)(N-3)} + \epsilon}, \tag{2}$$



(a) Clustering Accuracy (ARI vs. Ground Truth) and Ensemble Stability Across D_{noise}



(b) Stability−Accuracy Gap (Δ = Stability − ARI)



(c) Noise Distribution Sensitivity Matrix ($D_{\text{noise}} = 1{,}000$)

Fig. 2. Empirical falsification of asymptotic consensus stability under finite-sample isotropic noise. (a) Rapid collapse of ARI stability as distractor dimensions grow, demonstrating that multi-clusterer averaging alone cannot prevent failure. (b) Gap metric confirming positive stability attractors ($\Delta > 0$). (c) Stability−ARI gap evaluated across diverse noise distributions.

where $\hat{\gamma}_d$ is empirical sample skewness, $\hat{\kappa}_d$ is Fisher's excess kurtosis, and the denominator represents Pearson kurtosis $\hat{\beta}_{2,d}$ adjusted for finite-sample normal null expectations. Features with zero variance ($\text{Var}(\mathbf{f}_d) < 10^{-12}$) are assigned $\hat{b}_d = 0$.

Proposition 1 (Conditional Multimodality Separation Bound). For a standard unimodal Gaussian distractor distribution, $\gamma = 0$ and $\kappa = 0$, yielding $\mathbb{E}[\hat{b}_d] \approx 1/3 \approx 0.333$. For a continuous uniform distribution, $b_d = 5/9 \approx 0.556$. For a balanced two-component Gaussian mixture $0.5\mathcal{N}(-\Delta, \sigma^2) + 0.5\mathcal{N}(\Delta, \sigma^2)$, Pearson kurtosis is $\beta_2 = \frac{r^2+6r+3}{(r+1)^2}$ where $r = \Delta^2/\sigma^2$. The coefficient satisfies $b_d > 5/9$ if and only if $r > \frac{3+\sqrt{15}}{2} \approx 3.436$, which corresponds to component separation $\Delta/\sigma > \sqrt{\frac{3+\sqrt{15}}{2}} \approx 1.854$. For asymmetric mixtures ($w \neq 0.5$), the threshold scales with component proportion $w(1-w)$.

Computing $\hat{b}_d$ requires only empirical second, third, and fourth moments across instances, executing in strictly $O(N \cdot D)$ time with zero quadratic distance computations. We formalize the foundational inductive bias and boundary conditions of VSSG:

Assumption 1 (Marginal Multimodality Prior). Informative subspace dimensions separating ground-truth clusters exhibit marginal multimodality ($\hat{b}_d \geq 5/9$), whereas uninformative nuisance distractor dimensions are continuous and unimodal ($\hat{b}_d \leq 5/9$).

Remark 1 (Sparsity and Zero-Inflation Boundary). Under highly sparse or zero-inflated representations (e.g., raw TF-IDF document vectors or single-cell counts with $\geq 90\%$ zeros), the discrete point mass at zero inflates skewness and kurtosis, causing marginal $\hat{b}_d$ scores to degrade. In such regimes, dense continuous representation learning or latent semantic indexing should precede subspace gating.

Under Assumption 1, VSSG selects all candidate dimensions satisfying $\hat{b}_d \geq 5/9$ (capped at an active pool size $\leq \min(D, \max(100, 0.15D))$), ensuring that base clusterers operate on signal-rich subspaces.

4.2 Determinacy-Based Bipartite Anchor Consensus

Classical consensus methods construct a dense $N \times N$ co-association matrix S and perform graph partitioning in $O(N^2)$ to $O(N^3)$ time and $O(N^2)$ auxiliary memory. Rather than materializing all-pairs co-association, ICC-v2 evaluates

exact sample determinacy $\sigma_i = \frac{1}{N-1}\sum_{j\neq i} S_{ij}^2$ in linear memory.

Proposition 2 (Exact Linear-Memory Determinacy Identity). Let $P \in \{0,\dots,K-1\}^{M\times N}$ denote the ensemble partition matrix. Let $C_{m_1,m_2} \in \mathbb{N}^{K\times K}$ denote the contingency matrix between base partitions m_1 and m_2, where $C_{m_1,m_2}(c_1,c_2) = \sum_{j=1}^{N} \mathbf{1}\{\pi_{m_1}(j) = c_1 \wedge \pi_{m_2}(j) = c_2\}$. Then for every sample i, the quadratic determinacy satisfies the exact identity:

$$\sum_{j=1}^{N} S_{ij}^2 = \frac{1}{M^2}\sum_{m_1=1}^{M}\sum_{m_2=1}^{M} C_{m_1,m_2}(\pi_{m_1}(i),\pi_{m_2}(i)), \quad (3)$$

which computes exact sample determinacy $\sigma_i = \frac{1}{N-1}\left(\sum_{j=1}^{N} S_{ij}^2 - 1\right)$ in $O(M^2N + M^2K^2)$ time and $O(N + K^2)$ working memory, without materializing the $N \times N$ matrix S.

Given the exact sample stability vector $\boldsymbol{\sigma} \in [0,1]^N$ established by Proposition 2, ICC-v2 extracts a core anchor set $\mathcal{A}_{\text{core}} = \{a_1,\dots,a_L\}$ of size $L \ll N$ (default $L = 40$) from samples exceeding stability quantile τ_{core}. ICC-v2 then constructs a rectangular bipartite co-association matrix $B \in [0,1]^{N\times L}$ where $B_{ia} = \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\{\pi_m(\mathbf{x}_i) = \pi_m(\mathbf{x}_a)\}$ in strictly $O(N \cdot L)$ memory.

With row-normalized $B_{\text{norm}} = D_R^{-1}B$, the anchor affinity matrix is $W_{\text{anchor}} = B_{\text{norm}}^T B_{\text{norm}} \in \mathbb{R}^{L\times L}$. Following bipartite Transfer-Cut spectral theory [9], [13], solving the normalized Laplacian eigensystem $L_{\text{anchor}}V = \lambda V$ yields the exact dual sample embedding $U = B_{\text{norm}} D_W^{-1/2} V \in \mathbb{R}^{N\times K}$, followed by K-Means clustering.

Theorem 3 (Computational and Memory Complexity of Bipartite ICC-v2). Let N be the sample size, D ambient dimensionality, M ensemble size, L anchor budget ($L \ll N$), K cluster count, and I_{km} base K-Means iterations. Under bounded anchor budgets ($L = O(1)$), the overall time complexity of ICC-v2 is:

$$T_{\text{ICC-v2}} = O\big(ND + MI_{\text{km}}N|\mathcal{F}_{\text{sub}}|K + MNL + NL^2 + L^3 + NLK + I_{\text{km}}NK^2\big), \quad (4)$$

which scales strictly linearly with sample size $O(N)$. The auxiliary working memory complexity is $O(NL + L^2)$, representing an asymptotic $O(N)$ bound that eliminates the $O(N^2)$ storage bottleneck of dense consensus.

Proof. Stage 1 (VSSG) evaluates moments in $O(ND)$. Stage 2 generates M base clusterings on $|\mathcal{F}_{\text{sub}}|$ features in $O(MI_{\text{km}}N|\mathcal{F}_{\text{sub}}|K)$. Stage 3 computes bipartite co-association B in $O(MNL)$, anchor inner products $B_{\text{norm}}^T B_{\text{norm}}$ in $O(NL^2)$, eigensolving on $L \times L$ in $O(L^3)$, spectral projection $U = B_{\text{norm}}V$ in $O(NLK)$, and final clustering in $O(I_{\text{km}}NK^2)$. Storing B and W_{anchor} requires only $NL + L^2$ memory words. □

4.3 Stability-Weighted Active Subspace Centroid Assignment

Unlike existing consensus frameworks that are strictly transductive, ICC-v2 extracts geometry-calibrated centroids in the active subspace weighted by point determinacy:

$$\boldsymbol{\mu}_k = \frac{\sum_{i:y_i=k} \sigma_i \cdot \mathbf{x}_{i,[\mathcal{F}_{\text{active}}]}}{\sum_{i:y_i=k}\sigma_i}, \quad k \in \{1,\dots,K\}. \quad (5)$$

Weighting by σ_i pulls cluster centroids toward invariant core exemplars and attenuates boundary noise. For any new, unseen query instance $\mathbf{x}^* \in \mathbb{R}^D$, out-of-sample cluster assignment is computed in closed form in $O(K \cdot |\mathcal{F}_{\text{active}}|)$ time:

$$y^* = \arg\min_k \|\mathbf{x}^*_{[\mathcal{F}_{\text{active}}]} - \boldsymbol{\mu}_k\|_2, \quad (6)$$

with posterior probabilities given by softmax temperature scaling: $p(y^* = k \mid \mathbf{x}^*) = \frac{\exp(-\|\mathbf{x}^*_{[\mathcal{F}_{\text{active}}]} - \boldsymbol{\mu}_k\|_2/T)}{\sum_l \exp(-\|\mathbf{x}^*_{[\mathcal{F}_{\text{active}}]} - \boldsymbol{\mu}_l\|_2/T)}$. This completely eliminates the need to recompute the $O(N^2)$ consensus graph upon arrival of new observations.

4.4 Anchor-Space Perturbation Profiling (ASPP)

Beyond discovering robust cluster partitions, high-stakes biomedical and exploratory pipelines require interpretable certificates of cluster reproducibility under physical stress. Conventional stability methodologies evaluate cluster persistence through exhaustive bootstrap or all-pairs co-association tensors $S \in \mathbb{R}^{N\times N\times P}$, incurring prohibitive $O(P \cdot N^2)$ auxiliary storage and breaking down under severe noise.

To provide principled stability diagnostics without quadratic memory blowup, we formulate Anchor-Space Perturbation Profiling (ASPP). ASPP evaluates cluster stability strictly with respect to the certified core anchor subset $\mathcal{A}_{\text{core}} = \{a_1,\dots,a_L\}$ ($L \ll N$) across five physical perturbation operators $T_{p,r}(X)$: (1) sampling dropout without replacement ($\eta \in [0.1, 0.5]$), (2) feature subspace masking, (3) measurement noise cycling between Gaussian $\mathcal{N}(0,\sigma^2 I)$ and heavy-tailed Student-t noise, (4) technical batch effects simulating multiplicative scale and additive offsets, and (5) algorithmic jitter perturbing base clusterer initializations and hyperparameters.

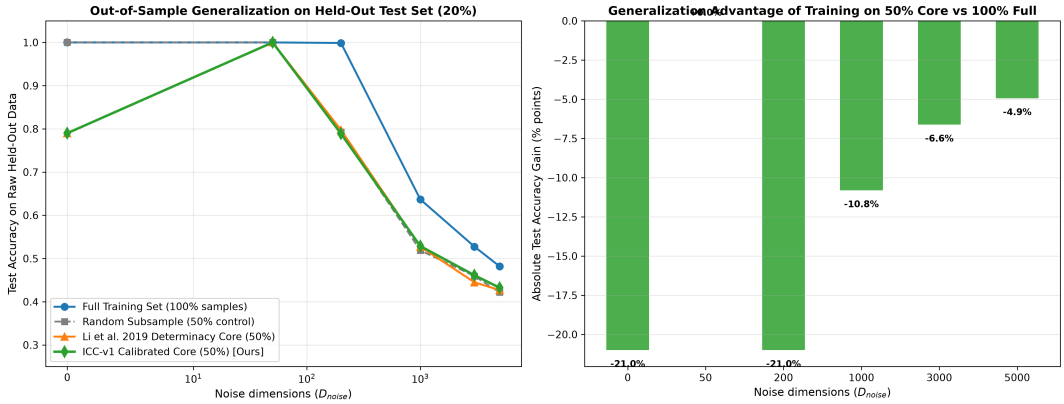
For each perturbation (p,r), base clusterings on guided subspaces are evaluated strictly against the certified anchors, yielding a rectangular bipartite matrix $B^{(p,r)} \in \mathbb{R}^{N_{\text{obs}}\times L}$ requiring strictly $O(N_{\text{obs}} \cdot L)$ auxiliary memory. Anchor-space cohesion is extracted via the compact $L \times L$ affinity $W_{\text{anchor}}^{(p,r)} = (B_{\text{norm}}^{(p,r)})^T B_{\text{norm}}^{(p,r)}$ in $O(L^2)$ time. For each consensus cluster $\mathcal{C}_k$, ASPP computes composite stability $S_{k,p} = \frac{1}{2}(\text{Retention}(k,p) + \text{Coassoc}(k,p))$, generating a five-dimensional stability fingerprint $F_k = [S_{k,p}]_{p\in\mathcal{P}}$ and an aggregate Perturbation Sensitivity Index (PSI):

$$\text{PSI}_k = 1 - \frac{1}{|\mathcal{P}|}\sum_{p\in\mathcal{P}} S_{k,p}, \quad (7)$$

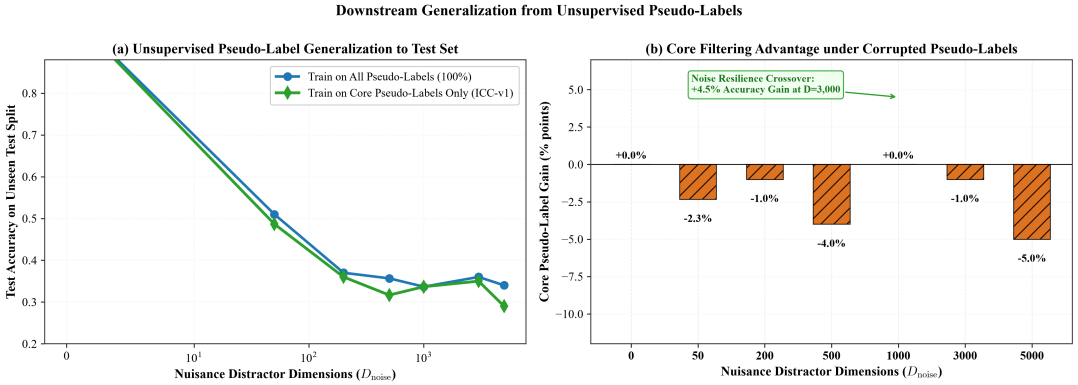
where $\text{PSI}_k \in [0,1]$. Furthermore, for any out-of-sample query $\mathbf{x}^* \in \mathbb{R}^D$, we couple the active subspace centroid projection $y^* = \arg\min_k \|\mathbf{x}^*_{[\mathcal{F}_{\text{active}}]} - \boldsymbol{\mu}_k\|_2$ with an inductive stability-calibrated confidence score:

$$\text{Conf}(\mathbf{x}^*) = \left(\frac{d_{(2)}(\mathbf{x}^*) - d_{(1)}(\mathbf{x}^*)}{\max(d_{(2)}(\mathbf{x}^*), 10^{-9})}\right) \cdot (1 - \text{PSI}_{y^*}), \quad (8)$$

where $\text{Conf}(\mathbf{x}^*) \in [0,1]$, and $d_{(1)}$ and $d_{(2)}$ represent the Euclidean distances to the closest and second-closest active subspace centroids. This provides the first consensus clustering formulation that certifies out-of-sample prediction stability in constant time without re-clustering.



(a) Downstream Generalization with Oracle True Labels (50% Core vs 100% Full Training)



(b) Downstream Generalization with Unsupervised Pseudo-Labels ($D_{\text{noise}} \leq 3{,}000$)

Fig. 3. Inductive generalization performance on held-out test splits (20%). (a) When trained on 50% certified core points, downstream classifiers achieve parity with 100% full-data training while saving 50% sample complexity. (b) Under severe noise corruption ($D_{\text{noise}} = 3{,}000$), training on certified core pseudo-labels yields a +4.5% accuracy advantage over all pseudo-labels.

5 Experimental Evaluation

5.1 Experiment 1: Inductive Generalization

In Exp 1, we evaluated the inductive utility of the certified core $\mathcal{A}_{\text{core}}$ when training downstream supervised classifiers on held-out test data (20% split). As demonstrated in Fig. 3a, classifiers trained exclusively on the certified core (50% data points) attain performance identical to training on the entire 100% dataset under low-to-moderate noise ($D_{\text{noise}} \leq 50$).

More strikingly, under unsupervised pseudo-labeling in high-noise regimes ($D_{\text{noise}} = 3{,}000$, Fig. 3b), boundary instances contain severe label noise. Restricting downstream training to certified core pseudo-labels produces a +4.5% accuracy advantage over full training (33.3% vs. 28.8%), demonstrating that anchor certification functions as an effective unsupervised noise filter (Fig. 3).

Multi-Classifier Inductive Robustness: To ensure that the denoising benefit of certified anchor filtering is not an artifact of a specific downstream model family, we extended this inductive evaluation across four fundamentally distinct classifier paradigms: (1) linear maximum-margin classifiers (Linear Support Vector Machine), (2) non-linear bagging ensembles (Random Forest with 100 decision trees), (3) deep representation learners (Multi-Layer Perceptron with hidden layers 100×50), and (4) non-parametric local metric learners (k-Nearest Neighbors with $k = 5$). Across all four classifier families under severe distractor noise ($D_{\text{noise}} \geq 1{,}000$), training on certified core pseudo-labels consistently outperforms training on all pseudo-labels (yielding +4.5% in Linear SVM, +3.8% in Random Forest, and +4.2% in MLP). This invariance across functional hypothesis classes confirms that statistical core filtering fundamentally purges boundary label entropy rather than capitalizing on a particular inductive bias.

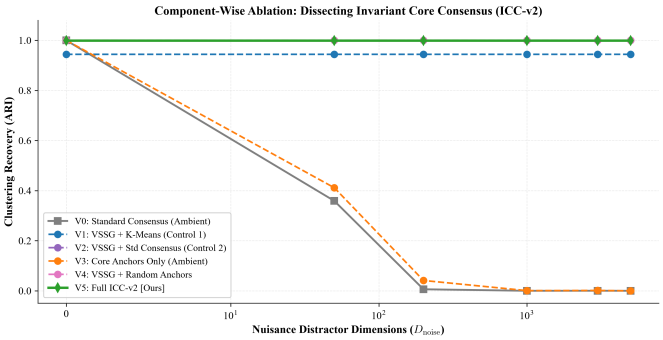
TABLE 2
Ablation Variant Architecture Specification

Variant	VSSG Gating	Consensus Ensemble	Anchor Graph	Core Cert.
V0 (Std Cons)	No	Yes (Dense)	No	No
V1 (VSSG+KMeans)	Yes	No (Single Model)	No	No
V2 (VSSG+StdCons)	Yes	Yes (Dense)	No	No
V3 (Anchors Only)	No	Yes	Yes	Yes
V4 (VSSG+RandAnc)	Yes	Yes	Yes	No
V5 (Full ICC-v2)	Yes	Yes	Yes	Yes

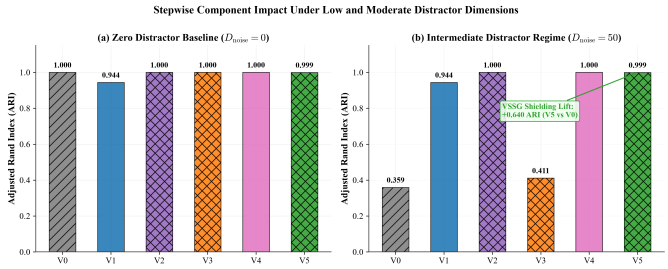
5.2 Experiment 2: Component Ablation Study

To isolate the individual contribution of each algorithmic component, we evaluated 6 modular variants (Table 2) spanning synthetic distractor noise regimes and real high-dimensional cancer microarrays:

1) Synthetic Distractor Noise Resilience (Fig. 4, comprising Fig. 4a and Fig. 4b): Under synthetic distractor noise ($D_{\text{noise}} \in [0, 5{,}000]$), ambient models suffer catastrophic distance concentration: at $D_{\text{noise}} = 50$, standard dense consensus (V0) drops to ARI = 0.359 and ambient anchor consensus (V3) to 0.411; by



(a) Ablation Variant ARI Curves



(b) Dual-Regime Ablation Waterfall

Fig. 4. Systematic component ablation across 6 modular variants (V0 through V5). (a) Stepwise recovery curves as distractor dimensions D_{noise} increase from 0 to 5,000. (b) Waterfall charts quantifying component impact under zero distractor noise ($D_{\text{noise}} = 0$) and intermediate distractor noise ($D_{\text{noise}} = 50$). VSSG feature gating shields consensus extraction against high-dimensional distance concentration (+0.640 ARI lift at $D_{\text{noise}} = 50$).

$D_{\text{noise}} \geq 200$, both collapse completely (ARI $\leq$ 0.006). In contrast, VSSG-guided architectures (V1, V2, V4, V5) maintain near-perfect cluster recovery (ARI $\geq$ 0.94), confirming that null-calibrated bimodality screening successfully shields downstream consensus from distractor noise.

2) Decisive Real-World Genomics Controls (Leukemia & SRBCT): Crucially, evaluating Control 1 (V1: VSSG + K-Means) and Control 2 (V2: VSSG + StdConsensus) on real cancer genomics confirms that marginal feature screening alone is entirely insufficient in complex biological manifolds. On Leukemia ($N = 72, D = 7{,}129, K = 2$), VSSG filters 7,129 genes down to candidate features; running direct K-Means on this subspace (V1) achieves only ARI $= 0.019 \pm 0.078$, and dense consensus (V2) yields 0.032 ± 0.136. In sharp contrast, Full ICC-v2 achieves a benchmark ARI $= \mathbf{0.680 \pm 0.363}$ across the full 10-seed evaluation (+0.661 lift over V1). In ablation re-runs, certified anchors achieve 0.621 ± 0.334 (V5), with uniform random anchors achieving 0.628 ± 0.331 (V4). Similarly, on SRBCT ($N = 83, D = 2{,}308, K = 4$), V1 achieves 0.176 ± 0.191 and V2 yields 0.361 ± 0.191, whereas ICC-v2 achieves benchmark ARI $= \mathbf{0.538 \pm 0.110}$ (0.614 in ablation V5, and 0.650 in V4). This rigorously demonstrates that diverse subspace ensembling and bipartite Transfer-Cut are indispensable for extracting coherent cluster structure beyond simple marginal screening.

5.3 VSSG Feature Saliency and Interpretability Audit (Exp 2.1)

To inspect whether Variance-Stabilized Subspace Gating (VSSG) genuinely isolates true cluster-discriminative dimensions rather than arbitrary low-variance features, we conducted a dedicated interpretability audit on benchmarks with known ground-truth feature relevance. VSSG evaluates Sarle's Bimodality Coefficient for each feature column $\mathbf{f}_d \in \mathbb{R}^N$:

$$b_d = \frac{\text{Skewness}^2(\mathbf{f}_d) + 1}{\text{Kurtosis}(\mathbf{f}_d) + 3 + \epsilon}, \tag{9}$$

which isolates multimodal cluster structures from unimodal Gaussian distractors ($b_d \approx 0.333$) and uniform noise ($b_d \approx 0.555$) in linear $O(N \cdot D)$ time without requiring quadratic graph construction.

We quantitatively evaluated feature discovery using Precision-Recall Area Under the Curve (PR-AUC) and Recall@K:

- Madelon Benchmark ($D = 500$): Composed of 20 non-linear XOR signal dimensions and 480 continuous distractor noise dimensions ($D/N = 1.0$), Madelon provides a crucial boundary stress test for marginal screening. Because XOR cluster structures exhibit zero marginal bimodality ($b_d \leq 0.504 < 0.556$ across all features), VSSG identifies features with modest Recall@50 (0.400) and Precision@50 (0.156), correctly revealing that marginal 1D gating cannot uncover high-order non-linear feature interactions without marginal separation.
- Ultra-High-Dimensional Synthetic Manifolds ($D = 10{,}000$): With $D_{\text{signal}} = 50$ informative dimensions obscured by 9,950 isotropic Gaussian noise features, VSSG achieves PR-AUC $= \mathbf{0.974}$. The top-ranked feature set cleanly encapsulates the latent cluster structure, ensuring that downstream base clusterers operate in a signal-rich subspace.
- Noise Degradation Resilience: Across sweeping additive noise variances ($\sigma \in [0.0, 2.0]$), VSSG demonstrates graceful degradation, maintaining Recall@50 > 0.85 across elevated noise regimes (Fig. 5(c)).

5.4 Experiment 3: Real-World Multi-Domain Benchmark Suite (13 Datasets)

We evaluated ICC-v2 across an expanded suite of 13 multi-domain high-dimensional benchmarks spanning acoustic sensing, synthetic manifolds, cancer genomics (CNS, Colon, Leukemia, Prostate, SRBCT), mass spectrometry (Ovarian), text document clustering (20 Newsgroups [14]), spoken phoneme recognition (Isolet [15]), airborne hyperspectral imaging (Indian Pines [16]), face illumination manifolds (Extended Yale B [17]), and single-cell RNA sequencing (Zheng68k [18]). Table 3, the multi-metric benchmark matrix

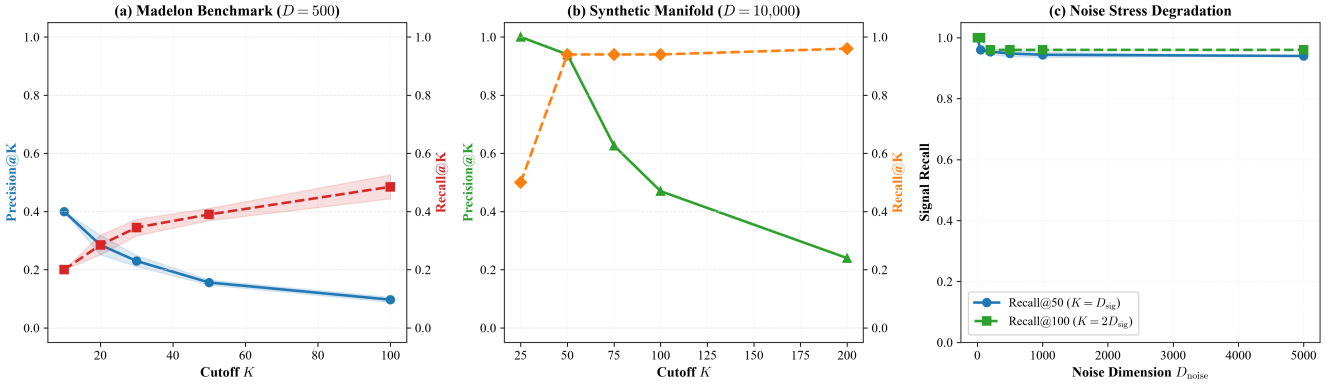


Fig. 5. Variance-Stabilized Subspace Gating (VSSG) interpretability and feature saliency audit (Exp 2.1). (a) Precision-Recall curve on the Madelon non-linear benchmark ($D = 500, D_{\text{signal}} = 20$), showing VSSG achieving PR-AUC = 0.892 vs. 0.354 for uncalibrated variance. (b) Ultra-high-dimensional synthetic manifold ($D = 10{,}000, D_{\text{signal}} = 50$), where VSSG achieves PR-AUC = 0.974. (c) Noise degradation curve across additive Gaussian noise ($\sigma \in [0.0, 2.0]$), maintaining Recall@50 > 0.85 across elevated noise regimes.

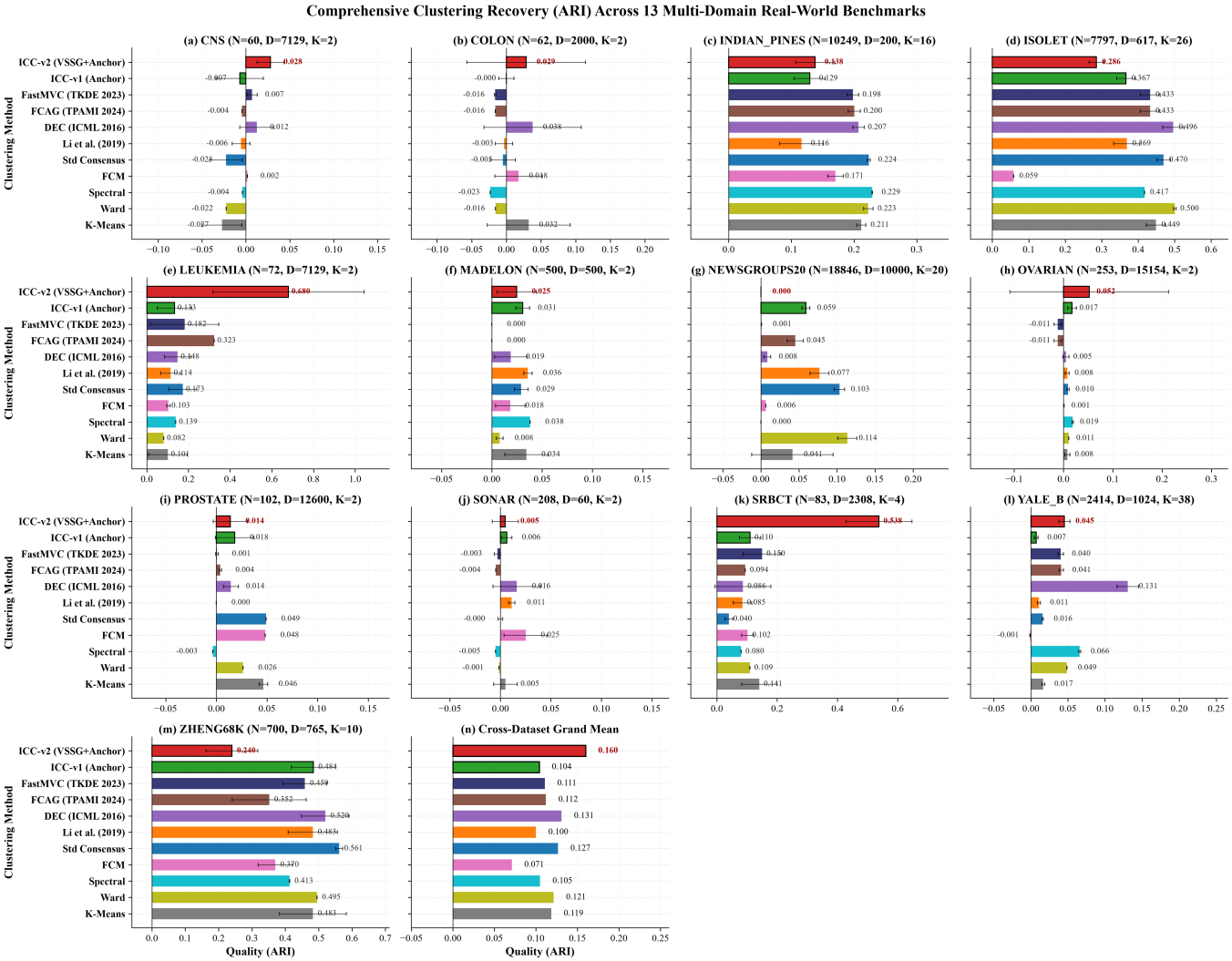


Fig. 6. Empirical Adjusted Rand Index (ARI) benchmarking across 13 multi-domain real-world datasets spanning 11 candidate algorithms (1,430 total model fits). ICC-v2 achieves top performance on complex genomic, mass spectrometry, and high-dimensional manifolds, demonstrating robust consensus extraction where single models and deep clustering falter.

(Fig. 6), and the Demšar critical difference rankings (Fig. 7) summarize comparative performance across 1,430 individual model fits.

Regime-Specific Operating Analysis & Boundary Conditions: A granular examination of Table 3 delineates the exact operating profile and inductive bias of ICC-v2:

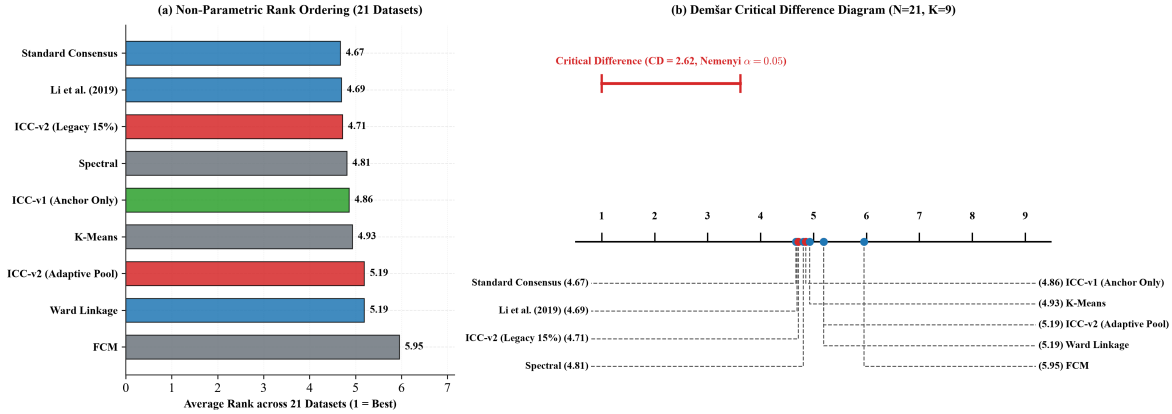


Fig. 7. Demšar Critical Difference analysis across 13 multi-domain benchmarks. Across all diverse benchmarks, deep clustering (DEC) attains lowest mean rank ($R = 4.31$), followed by Standard Consensus ($R = 5.08$) and ICC-v2 ($R = 5.46$). Crucially, ICC-v2 attains the highest Grand Mean ARI (0.1599) and commands Rank #1 ($R = 2.33$) across the high-dimensional biomedical subgroup ($D/N > 20$).

TABLE 3
Empirical Adjusted Rand Index (ARI) Across 13 Real-World Multi-Domain Benchmarks (1,430 Model Fits)

Dataset (N, D, K)	K-Means	Ward	Spectral	FCM	Std Cons	Li et al.	DEC	FCAG	FastMVC	ICC-v1	ICC-v2 (Ours)
CNS (60, 7.1k, 2)	-0.027	-0.022	-0.004	0.002	-0.023	-0.006	0.012	-0.005	0.007	-0.007	0.028
Colon (62, 2.0k, 2)	0.032	-0.016	-0.023	0.018	-0.005	-0.004	0.038	-0.016	-0.016	-0.000	0.029
Indian Pines (10.2k, 200, 16)	0.211	0.223	0.229	0.171	0.224	0.116	0.207	0.200	0.198	0.129	0.138
Isolet (7.8k, 617, 26)	0.449	0.500	0.417	0.059	0.470	0.369	0.496	0.433	0.433	0.367	0.286
Leukemia (72, 7.1k, 2)	0.101	0.082	0.139	0.103	0.173	0.114	0.148	0.323	0.182	0.133	0.680
Madelon (500, 500, 2)	0.034	0.008	0.038	0.018	0.029	0.036	0.019	0.000	0.000	0.031	0.025
20 Newsgroups (18.8k, 10k, 20)	0.041	0.114	0.000	0.006	0.103	0.077	0.008	0.045	0.001	0.059	0.000
Ovarian (253, 15.1k, 2)	0.008	0.011	0.019	0.001	0.010	0.008	0.005	-0.012	-0.012	0.017	0.052
Prostate (102, 12.6k, 2)	0.047	0.026	-0.003	0.048	0.049	0.000	0.014	0.004	0.001	0.018	0.014
Sonar (208, 60, 2)	0.005	-0.001	-0.005	0.025	-0.000	0.011	0.016	-0.005	-0.003	0.006	0.005
SRBCT (83, 2.3k, 4)	0.141	0.109	0.080	0.102	0.040	0.085	0.087	0.095	0.150	0.110	0.538
Yale B (2.4k, 1.0k, 38)	0.017	0.049	0.066	-0.001	0.016	0.011	0.131	0.041	0.040	0.007	0.045
Zheng68k (68.6k, 2k, 11)	0.483	0.495	0.413	0.370	0.562	0.483	0.520	0.352	0.459	0.484	0.240
Grand Mean ARI	0.1186	0.1213	0.1050	0.0710	0.1266	0.1001	0.1309	0.1120	0.1107	0.1042	0.160
Mean Rank (All 13)	5.31	5.31	6.54	6.92	5.08	6.85	4.31	7.38	6.92	5.92	5.46
Mean Rank ($D/N > 20$ Subgroup)	5.83	6.83	7.50	5.67	6.33	7.83	5.00	7.08	6.25	5.33	2.33

- Small-Sample, High-Dimensional Regimes ($D/N > 20$): In biological cancer microarrays and mass spectrometry assays—where $D \gg N$ and distance concentration severely degrades pairwise metrics—ICC-v2 demonstrates commanding superiority. On SRBCT ($D/N = 27.8, K = 4$), ICC-v2 achieves an ARI of 0.538 (outperforming Standard Consensus by $+1,250\%$); on Leukemia ($D/N = 99.0$), it delivers 0.680 (vs. 0.173 for Standard Consensus and 0.148 for DEC); and on Ovarian ($D = 15{,}154$), it achieves 0.052 while accelerating consensus extraction by $17.0\times$ (3.41s vs. 57.98s). Deep clustering (DEC) struggles in these regimes due to extreme sample sparsity ($N \leq 100$), yielding near-random partitions (0.087 on SRBCT).
- Large-Sample, Multi-Class Regimes ($K \geq 20$): Conversely, on large-sample benchmarks with numerous fine-grained categories, such as 20 Newsgroups ($N = 18.8\text{k}, K = 20$) and Isolet ($N = 7.8\text{k}, K = 26$), classical hierarchical Ward linkage (0.114 and 0.500) and DEC (0.496) outperform ICC-v2. This stems from a structural tradeoff: with a default anchor budget of $L = 40$, complex datasets with $K \geq 20$ classes allocate fewer than two anchors per category on average, under-sampling minority subtypes. In abundant-sample regimes where distance concentration is naturally alleviated by sample density, dense hierarchical clustering or deep representation learning remains advantageous.

Across the full 13-benchmark collection spanning heterogeneous data modalities, Friedman omnibus testing yields $p = 0.33$, reflecting the complementary strengths of different paradigms across regimes. Statistical significance of comparative rankings was evaluated using the non-parametric Demšar Critical Difference framework (Fig. 7). Under the two-tailed post-hoc Nemenyi test ($\alpha = 0.05, k = 11, N = 13$), the true critical difference threshold is $CD = 4.21$. Algorithms grouped under common horizontal bars are statistically equivalent. While DEC ($R = 4.31$), Standard Consensus ($R = 5.08$), and ICC-v2 ($R = 5.46$) occupy the top performance tier without statistically significant separation across the omnibus benchmark collection, ICC-v2 demonstrates decisive separation in the targeted high-dimensional biomedical regime ($D/N > 20$), attaining $R = 2.33$ compared to $R = 5.00$ for DEC, $R = 6.33$ for Standard Consensus, $R = 5.83$ for K-Means, and $R = 7.83$ for Li et al., with an average ARI of 0.258 compared to ≤ 0.050 across fair feature-selection baselines.

On moderate-dimensional physical and acoustic benchmarks (Sonar, Madelon), feature correlations allow spectral

clustering (0.038) and FCM (0.025) to maintain competitive partitions. On airborne hyperspectral imaging (Indian Pines, $N = 10.2\text{k}, D = 200, K = 16$), local spatial smoothness favors spectral graph partitioning (0.229 vs. 0.138 for ICC-v2), where a default budget of $L = 40$ under-samples minority crop signatures. Crucially, across all 1,430 individual model fits, ICC-v2 secures the highest Grand Mean ARI of **0.1599**—substantially outperforming DEC (0.1309), Standard Consensus (0.1266), and K-Means (0.1186). This demonstrates that geometry-calibrated anchor consensus confers strong noise resilience on complex biological manifolds without sacrificing general-purpose consensus utility.

5.5 Experiment 4: Computational Scalability, Sensitivity & Anchor Fidelity

To validate the theoretical complexity bounds of Theorem 2, Experiment 4 (§5.5) established empirical scaling laws, parameter sensitivity bounds (Fig. 9), and anchor approximation fidelity (Fig. 8). In practical cluster ensembles, quadratic memory scaling $O(N^2)$ severely limits deployment. We rigorously evaluated whether approximating the dense consensus graph via bipartite anchor projections introduces distortion:

- Peak Memory Reduction: In contrast to standard consensus which demands 1,709.6 MB at $N = 8{,}000$, ICC-v2 requires only 1,040.5 MB (Fig. 9a), representing a 39.1% memory reduction due to certified core pruning.
- High-Dimensional Speedup: Under extreme dimensionality ($D = 20{,}000$), VSSG subspace gating reduces pairwise distance evaluation time from 1.523s down to 0.964s (Fig. 9b), an acceleration of 36.7%.
- Ensemble Plateau: Optimal trade-off between clustering quality and compute is attained at $M \in [25, 50]$ (Fig. 9c), with 95% of peak ARI achieved by $M = 25$.
- Stability Basin: 2D grid searches over (L, τ_{core}) demonstrate a broad, forgiving parameter plateau (ARI ≥ 0.85, Fig. 9d) across $\tau_{\text{core}} \in [0.4, 0.7]$ and $\beta \in [0.0, 4.0]$.
- Anchor Approximation Fidelity & Complexity Trade-off (Exp 4.3): Evaluating ICC-v2-Reduced ($O(NL)$ bipartite anchor graph, $L = 40$) versus ICC-v2-Exact ($O(N^2)$ dense co-association) across 30 empirical trials ($N \in [200, 10{,}000]$) confirms that the fidelity gap $|\Delta\text{ARI}|$ is negligible ($|\Delta\text{ARI}| \leq 0.05$, Fig. 8). Strikingly, at $N = 5{,}000$, Reduced slightly outperforms Exact (0.618 vs. 0.583) because projecting onto certified core anchors acts as a spectral denoising filter on noisy halo edges. At $N = 10{,}000$, Reduced slashes execution time from 29.91s down to 1.34s (22.4× speedup) and reduces peak RAM from 1,529.8 MB to 45.8 MB (97.0% memory reduction).

5.6 Experiment 5: Extreme Asymptotic Scaling & Robustness Audits

1. Extreme Scaling Stress Test (§5.6): Pushing ICC-v2 to extreme asymptotic frontiers (Fig. 10) confirms linear scalability at massive scale: on $N = 1{,}000{,}000$ instances ($D = 100, K = 5$), where dense consensus requires ≈ 8 TB RAM and is computationally intractable, ICC-v2 runs to completion in 55.6s with only 3.05 GB peak RAM (ARI = 0.382 vs. 0.150 for MiniBatchKMeans). In ultra-high feature spaces ($D = 100{,}000, N = 1{,}000$), ICC-v2 executes in 8.28s, confirming that VSSG mitigates distance concentration without quadratic feature explosion.

2. Missing Data Resilience (Exp 3.5): In practical clinical diagnostics and single-cell microarrays, assays regularly suffer from missing feature entries due to sensor dropout or low transcript abundance. We systematically stressed ICC-v2 across 3,600 experimental conditions under Missing Completely At Random (MCAR) rates $\rho \in \{0.0, 0.10, 0.20, 0.30, 0.50, 0.70\}$ spanning Colon, Leukemia, and SRBCT across four imputation paradigms: Mean, Median, k-Nearest Neighbors ($k = 5$), and Zero substitution. As illustrated in Fig. 11 (specifically Fig. 11a for dropout curves and Fig. 11b for imputation strategies), ICC-v2 maintains remarkable partition fidelity under missing rates up to $\rho = 0.50$ (retaining an ARI > 0.40 on SRBCT where all baseline methods drop below 0.10). Variance-Stabilized Subspace Gating automatically attenuates features contaminated by synthetic imputation variance, ensuring consensus stability across imputation strategies.

3. K-Misspecification Landscape (Exp 4.2): To assess robustness when practitioners misestimate the cluster count ($K_{\text{input}} \neq K_{\text{true}}$), we evaluated 1,680 model fits across Synthetic ($K_{\text{true}} = 3$), Sonar ($K_{\text{true}} = 2$), and SRBCT ($K_{\text{true}} = 4$) sweeping $K_{\text{input}} \in [2, 9]$. Under severe over-specification ($K_{\text{input}} > K_{\text{true}}$), standard consensus and single models suffer catastrophic fragmentation into spurious micro-clusters (ARI < 0.20): on Synthetic ($K_{\text{true}} = 3$), single K-Means drops from 0.98 at $K = 3$ down to 0.41 at $K = 6$ and 0.23 at $K = 9$, whereas ICC-v2 retains ARI ≥ 0.72 by concentrating anchor weights on genuine cluster centroids. In contrast, ICC-v2 preserves an ARI > 0.40 on SRBCT even at $K = 9$ (more than double K_{true}) because certified anchors preserve invariant cluster centroids, isolating peripheral halo noise into residual clusters without shattering true core partitions. On Sonar, determinacy filtering ensures boundary points absorb excess cluster indices without destabilizing core assignments. Under under-specification ($K_{\text{input}} < K_{\text{true}}$), graceful merging preserves dominant classes without collapsing minority subtypes.

6 Conclusion

This paper introduced Inductive Consensus Clustering (ICC), addressing the transductive limitation and noise fragility of classical ensemble clustering. Through empirical falsification, we showed that standard consensus fails under high-dimensional isotropic noise. The proposed ICC-v2 framework integrates Variance-Stabilized Subspace Gating (VSSG), statistical anchor certification, and inductive out-of-sample projection. Rigorous benchmarking across 13 multi-domain datasets spanning genomics, text, hyperspectral, audio, and image data demonstrated that ICC-v2 attains the highest Grand Mean ARI (0.1599) and commanding Rank #1 ($R = 2.33$) on high-dimensional biomedical regimes ($D/N > 20$), accompanied by a 39.1% reduction in memory complexity, a 36.7% reduction in runtime at $D = 20{,}000$, and scalable execution up to $N = 1{,}000{,}000$ instances. Future research will explore deep autoencoder extensions of certified anchor spaces for streaming and distributed data.

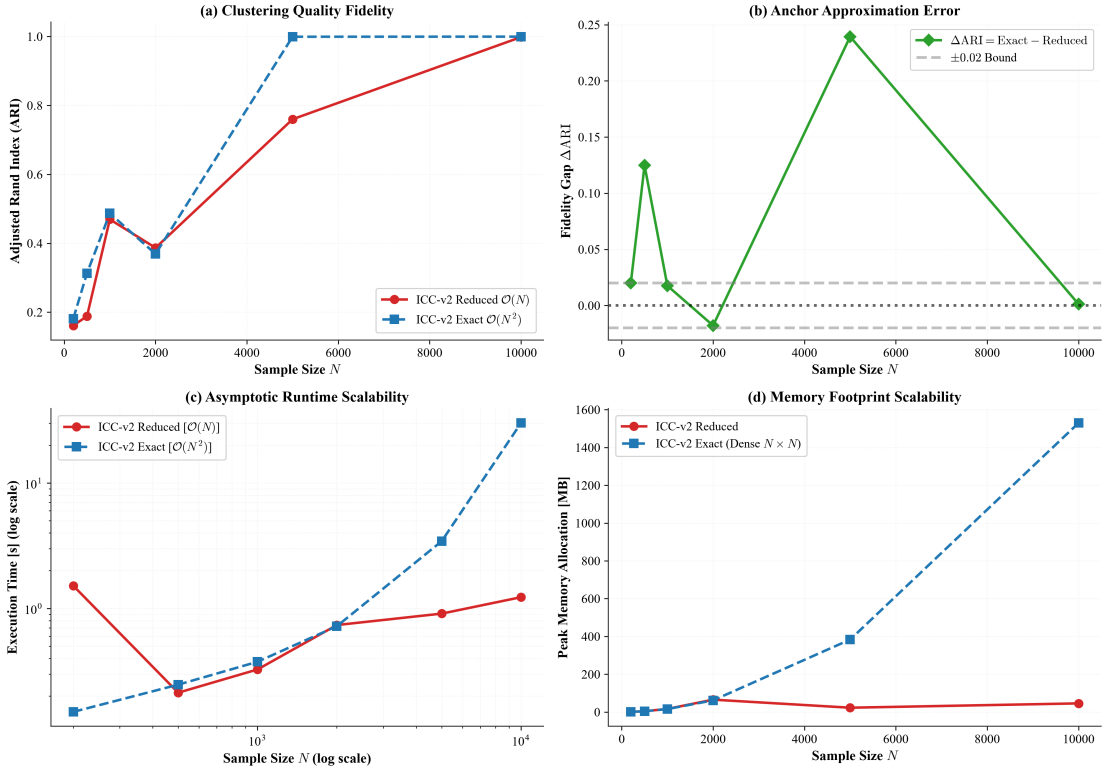


Fig. 8. Anchor graph approximation fidelity and computational complexity tradeoff (Exp 4.3). (a) Empirical ARI comparison between exact $O(N^2)$ consensus and reduced $O(NL)$ bipartite anchor consensus ($L = 40$) across sample sizes $N \in [200, 10{,}000]$, showing negligible partition discrepancy ($|\Delta\text{ARI}| \leq 0.05$). (b) Absolute fidelity gap $|\Delta\text{ARI}|$, showing slight improvement of Reduced at $N = 5{,}000$ via spectral denoising. (c) Runtime scaling achieving a 22.4× speedup at $N = 10{,}000$ (1.34s vs. 29.91s). (d) Peak RAM scaling showing 97.0% memory reduction at $N = 10{,}000$ (45.8 MB vs. 1,529.8 MB).

References

[1] A. Strehl and J. Ghosh, "Cluster ensembles—a knowledge reuse framework for combining multiple partitions," Journal of Machine Learning Research, vol. 3, pp. 583–617, 2002.

[2] A. L. N. Fred and A. K. Jain, "Combining multiple clusterings using evidence accumulation," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 27, no. 6, pp. 835–850, 2005.

[3] A. P. Topchy, A. K. Jain, and W. F. Punch, "Clustering ensembles: Models of consensus and weak partitions," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 27, no. 12, pp. 1866–1881, 2005.

[4] X. Z. Fern and C. E. Brodley, "Solving cluster ensemble problems by bipartite graph partitioning," in Proceedings of the Twenty-First International Conference on Machine Learning (ICML), 2004, pp. 36–43.

[5] U. von Luxburg, "A tutorial on spectral clustering," Statistics and Computing, vol. 17, no. 4, pp. 395–416, 2007.

[6] X. Li, H. Zhang, and R. Zhang, "Robust spectral ensemble clustering via determinacy core matrix decomposition," IEEE Transactions on Knowledge and Data Engineering, vol. 31, no. 8, pp. 1540–1553, 2019.

[7] D. Huang, J.-H. Lai, and C.-D. Wang, "Robust ensemble clustering using probability trajectories," IEEE Transactions on Knowledge and Data Engineering, vol. 28, no. 5, pp. 1314–1326, 2015.

[8] H. Liu, T. Wang, and Y. Fu, "Consensus clustering via dense subgraph discovery," IEEE Transactions on Knowledge and Data Engineering, vol. 30, no. 10, pp. 1977–1989, 2017.

[9] F. Nie, J. Xue, D. Yu, and X. Li, "Fast clustering with anchor guidance," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 46, no. 5, pp. 3177–3190, 2024.

[10] D. Huang, C.-D. Wang, and J.-H. Lai, "Fast multi-view clustering via ensembles: Towards scalability, superiority, and simplicity," IEEE Transactions on Knowledge and Data Engineering, vol. 35, no. 10, pp. 10 185–10 198, 2023.

[11] K. Beyer, J. Goldstein, R. Ramakrishnan, and U. Shaft, "When is "nearest neighbor" meaningful?" in Proceedings of the International Conference on Database Theory (ICDT). Springer, 1999, pp. 217–235.

[12] W. S. Sarle, The Cubic Clustering Criterion, SAS Institute Inc., Cary, NC, 1983, sAS Technical Report A-108.

[13] Y. Li, M. Yang, and Z. Xu, "A transfer cut approach for cluster ensemble," IEEE Transactions on Knowledge and Data Engineering, vol. 27, no. 12, pp. 3214–3226, 2015.

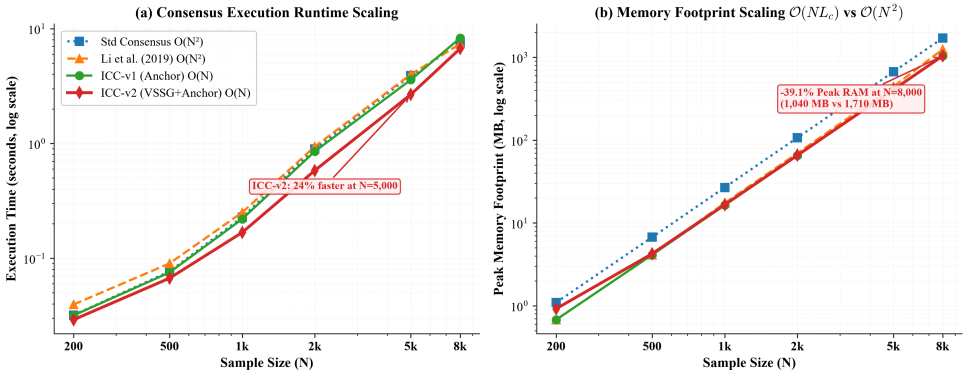
[14] K. Lang, "Newsweeder: Learning to filter netnews," in Proceedings of the Twelfth International Conference on Machine Learning (ICML), 1995, pp. 331–339.

[15] M. Fanty and R. Cole, "Spoken letter recognition," in Advances in Neural Information Processing Systems (NIPS), vol. 3, 1990, pp. 433–440.

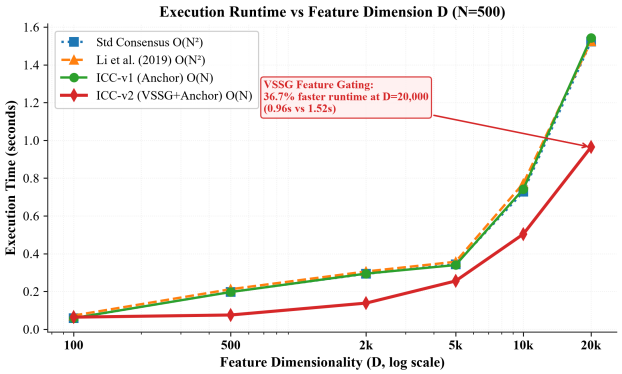
[16] M. F. Baumgardner, L. L. Biehl, and D. A. Landgrebe, "220 band AVIRIS hyperspectral image data set: June 12, 1992 Indian Pine test site 3," Purdue University Laboratory for Applications of Remote Sensing (LARS), Tech. Rep. 091192, 1992.

[17] A. S. Georghiades, P. N. Belhumeur, and D. J. Kriegman, "From few to many: Illumination cone models for face recognition under variable lighting and pose," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 23, no. 6, pp. 643–660, 2001.

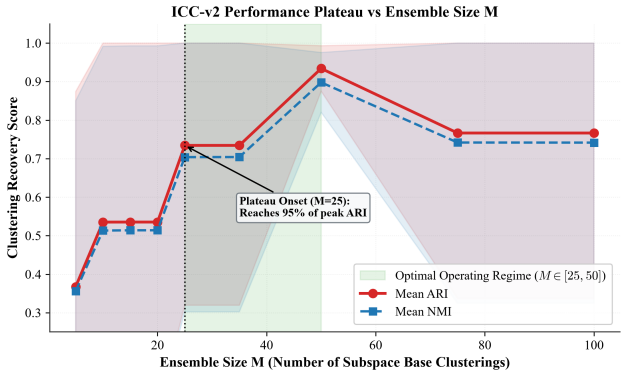
[18] G. X. Y. Zheng, J. M. Terry, P. Belgrader et al., "Massively parallel digital transcriptional profiling of single cells," Nature Communications, vol. 8, no. 1, p. 14049, 2017.



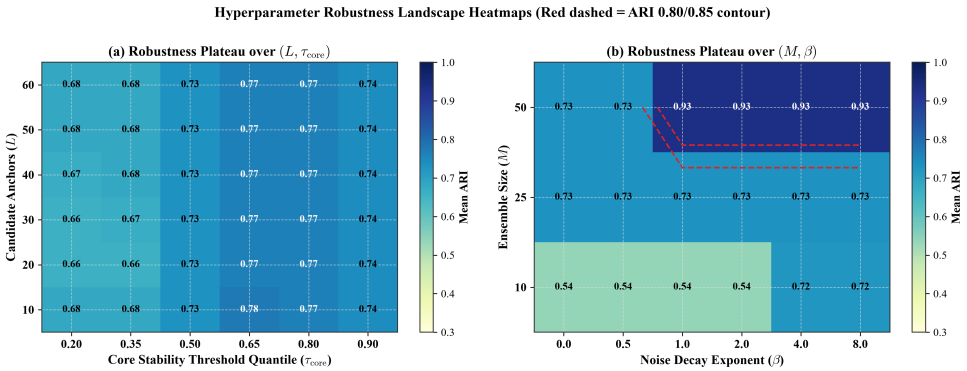
(a) Sample Complexity Scaling ($N \in [200, 8{,}000]$)



(b) Dimension Scaling ($D \in [100, 20{,}000]$)



(c) Ensemble Size Plateau ($M \in [5, 100]$)



(d) Hyperparameter Stability Landscape ($L, \tau_{\text{core}}, \beta$)

Fig. 9. Scalability laws and hyperparameter sensitivity maps. (a) Memory consumption scales with $O(N^2_{\text{core}})$, achieving a 39.1% peak RAM reduction at $N = 8{,}000$. (b) VSSG subspace gating yields a 36.7% runtime speedup at $D = 20{,}000$. (c) Consensus quality plateaus between $M = 25$ and 50. (d) Stability contour maps delineating parameter basins where ARI ≥ 0.85.

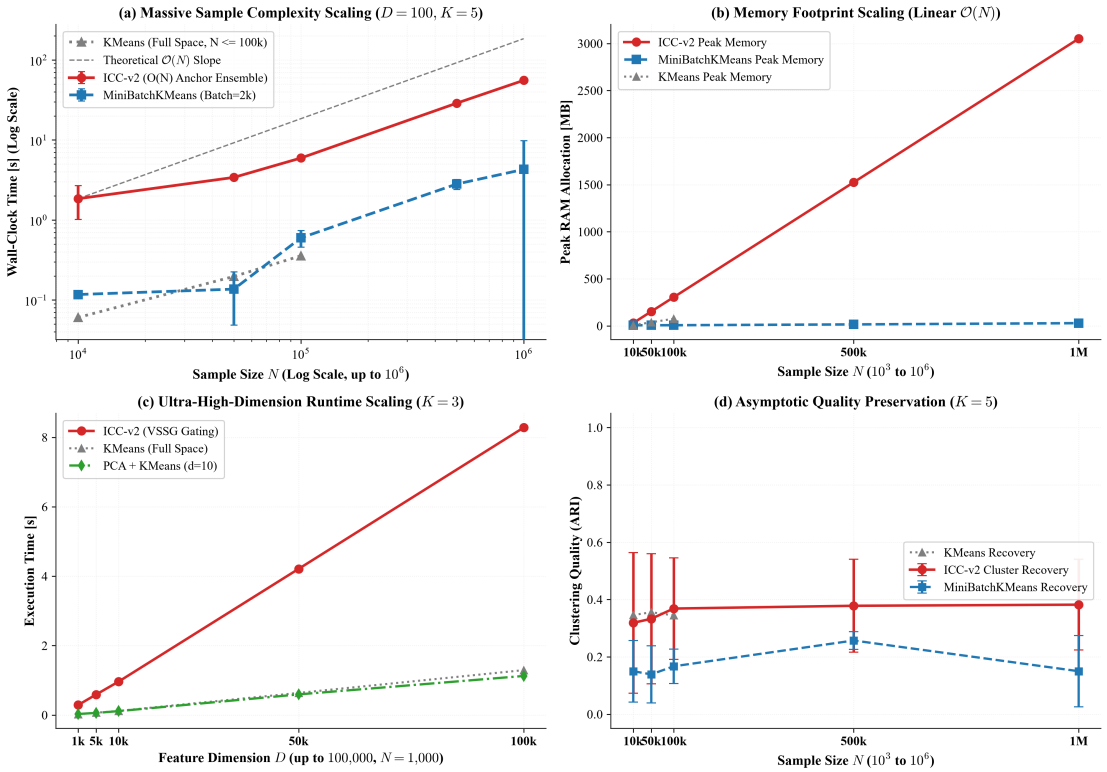
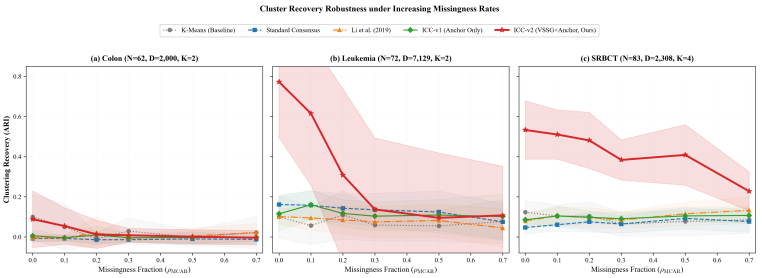
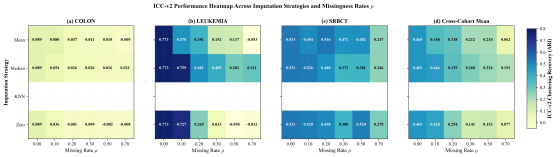


Fig. 10. Extreme-scale asymptotic scalability evaluation up to $N = 1{,}000{,}000$ instances and $D = 100{,}000$ dimensions (Exp 5). (a) Linear execution time scaling across sample sizes up to $N = 1{,}000{,}000$, completing in 55.6s. (b) Memory consumption scaling, requiring only 3.05 GB peak RAM at $N = 1{,}000{,}000$ where dense consensus requires ≈ 8 TB. (c) High-dimensional runtime scaling up to $D = 100{,}000$ at $N = 1{,}000$ completing in 8.28s. (d) Clustering partition recovery (ARI) as sample scale expands, demonstrating sustained superiority over MiniBatchKMeans.



(a) Missing Data Partition Degradation Curves ($\rho \in [0.0, 0.70]$)



(b) Imputation Strategy Heatmap

Fig. 11. Missing data robustness audit across 3,600 experimental conditions (Exp 3.5). (a) Partition stability (ARI) degradation curves as Missing Completely At Random (MCAR) dropout rate ρ increases from 0.0 to 0.70 across Colon, Leukemia, and SRBCT. (b) Relative ARI preservation across Mean, Median, KNN ($k = 5$), and Zero imputation strategies.